Appendix. A flood without end: the homotopy type of a monotone union

Milnor, "Morse Theory", Appendix, rewritten for a first-year graduate student who wants to know what the theorem is for. Every proof is complete; Whitehead's theorem and standard facts about CW-complexes are quoted with references.

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0. A flood that never stops rising

Let water rise slowly over a landscape. At level a the flooded region is M^a, the points of height at most a, and Chapter 2 (Milnor §3) describes what happens when the level passes a critical height: a cell is attached for each pass. With finitely many passes, the shape of the flooded land is known once the water is above the highest one. In the landscape that matters most in this book there are infinitely many passes. The landscape is the space $\Omega = \Omega(M; p, q)$ of paths from p to q in a complete Riemannian manifold, the height is the energy E, and the passes are the geodesics from p to q. On a sphere there are infinitely many of them, with energies tending to infinity (Chapter 7, §6): the water never stops rising. Chapter 10 shows that each flooded region Ω^a has the homotopy type of a finite CW-complex, with one cell for each geodesic of energy below a. The question of this appendix is:

If we know the shape, up to homotopy, of the flooded region at every finite level, do we know the shape of the whole space?

The answer is yes when every point is under water *with some margin* at some finite level. Milnor's Example 3 shows how it fails otherwise: a point that the water reaches only at the edge can come apart from the rest. The tool is the *telescope*, which records the whole history of the flood at once, and the main result, Theorem A, says that maps which are homotopy equivalences level by level give a homotopy equivalence of the whole. The application (§7) assembles the space of paths on a sphere from its energy levels and reads off its homology.

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1. Telescopes (Milnor p. 149)

Milnor's purpose is an alternative to the last step in the proof of Theorem 17.3: given $\Omega^{a_0} \subseteq \Omega^{a_1} \subseteq \cdots$ in Ω, each of the homotopy type of a CW-complex, to show that the union Ω has the homotopy type of a CW-complex. More generally, let X be a topological space and $X_0 \subseteq X_1 \subseteq X_2 \subseteq \cdots$ subspaces. To what extent is the homotopy type of X determined by those of the X_i?

Definition. The *telescope* of the sequence is

$$X_\Sigma = X_0 \times [0,1] \cup X_1 \times [1,2] \cup X_2 \times [2,3] \cup \cdots \subseteq X \times \mathbb{R},$$

with the subspace topology. X is the *homotopy direct limit* of $\{X_i\}$ if the projection $p : X_\Sigma \to X$, $p(x,\tau) = x$, is a homotopy equivalence.

Since $X_{n-1} \subseteq X_n$, a pair (x,τ) with $\tau \geq 0$ lies in X_Σ exactly when $x \in X_{\lfloor \tau \rfloor}$. Three elementary facts are used throughout, by Milnor without comment.

(T1) *Upward closure.* If $(x,\tau) \in X_\Sigma$ and $\tau' \geq \tau$, then $(x,\tau') \in X_\Sigma$. So a homotopy that only increases the second coordinate stays in X_Σ.

(T2) *Truncations.* The truncation $X_\Sigma^n = \{(x,\tau) \in X_\Sigma : \tau \le n\} = X_0 \times [0,1] \cup \cdots \cup X_{n-1} \times [n-1, n] \cup X_n \times \{n\}$ deformation retracts onto $X_n \times \{n\}$ by $(x,\tau) \mapsto (x, \tau + s(n-\tau))$, $0 \le s \le 1$, by (T1) and $X_{\lfloor \tau \rfloor} \subseteq X_n$. Hence $x \mapsto (x,n)$ is a homotopy equivalence $X_n \to X_\Sigma^n$.

(T3) *Pasting.* If every X_i is closed in X, the strips $X_n \times [n, n+1]$ are closed in X_Σ and form a locally finite family (the open set $\{|\tau - \tau_0| < \frac{1}{2}\}$ meets at most two). So a map on X_Σ, or on $X_\Sigma \times [0,1]$, is continuous as soon as it is continuous on each strip (times $[0,1]$).

> **A hidden hypothesis.** Milnor's proof of Theorem A defines maps strip by strip, and (T3) is what makes such a map continuous. Case 1 of his proof (§4), as written, needs the X_i closed. Take $X = [0,1]$, $X_0 = (0,1]$, $X_n = X$ for $n \ge 1$, $f = \mathrm{id}$, and for the homotopy h_u^0 of §4 from the identity of $(0,1]$ to itself choose $h_u^0(x) = x + (1-x)\max\big(0, 1 - |u - 1 + \frac{x}{2}|/\frac{x}{2}\big)$, which pushes x all the way to 1 at the time $u = 1 - \frac{x}{2}$. Then the map g of §4 is discontinuous at $(0,1)$. Case 2 (§5) applies Case 1 only to truncations of telescopes, which are always closed, and the one map it defines strip by strip on a telescope of the original subspaces is $G : Y_\Sigma \to X_\Sigma$; so Milnor's proof of Theorem A needs only the Y_i closed. In his application they are subcomplexes, hence closed. Theorem A itself needs no closedness at all: at the end of §5 Milnor's proof is run on mapping telescopes, where a map given block by block is always continuous.

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2. Milnor's three examples (Milnor pp. 149–150)

Example 1. Suppose that each point of X lies in the interior of some X_i, and that X is paracompact. Then X is the homotopy direct limit of $\{X_i\}$.

Milnor's proof: a partition of unity gives a map $f : X \to \mathbb{R}$ with $f \ge 0$ and $f(x) \ge i+1$ for $x \notin X_i$, and $x \mapsto (x, f(x))$ embeds X in X_Σ as a subset that is "clearly a deformation retract".

Construction of f. A paracompact (Hausdorff) space has a partition of unity subordinate to any open cover (Munkres, *Topology*, 2nd ed., Theorem 41.7). Take $\{\varphi_i\}_{i \ge 0}$ with $\operatorname{supp} \varphi_i \subseteq \operatorname{int} X_i$ and put $f = \sum_i i\,\varphi_i$, continuous as a locally finite sum, with $f \ge 0$. If $x \notin X_i$ then $x \notin X_j$ for $j \le i$, so $\varphi_j(x) = 0$ for $j \le i$ and $f(x) = \sum_{j>i} j\,\varphi_j(x) \ge i+1$. The rest is the following lemma.

Lemma 2.1. If $f : X \to [0,\infty)$ is continuous and $f(x) \ge i+1$ whenever $x \notin X_i$, then $p : X_\Sigma \to X$ is a homotopy equivalence.

Proof. For $x \in X$ let $m = \lfloor f(x) \rfloor$. As $f(x) < m+1$, $x \in X_m$, so $\sigma(x) = (x, f(x)) \in X_\Sigma$. The map $\sigma : X \to X_\Sigma$ is continuous with $p\sigma = \mathrm{id}_X$, hence a homeomorphism onto the graph Γ of f. The homotopy

$$H_s(x,\tau) = \big(x, (1-s)\tau + s f(x)\big), \qquad 0 \le s \le 1,$$

stays in X_Σ: if $\tau \le f(x)$ the second coordinate only increases (T1); if $\tau \ge f(x)$ it stays $\ge f(x)$, and $(x,\tau') \in X_\Sigma$ for $\tau' \ge f(x)$ by (T1) applied to $\sigma(x)$. Since $H_0 = \mathrm{id}$, $H_1 = \sigma p$ and each H_s fixes Γ, this is a deformation retraction onto Γ, and $\sigma p \simeq \mathrm{id}$, $p\sigma = \mathrm{id}$. □

Lemma 2.1 needs no paracompactness once such an f is at hand. In particular:

Lemma 2.2. X_Σ is the homotopy direct limit of its truncations $\{X_\Sigma^n\}$.

Proof. Take $f(x,\tau) = \tau + 1$: if $(x,\tau) \notin X_\Sigma^n$, then $\tau > n$ and $f > n+1$. □

Milnor needs Lemma 2.2, without saying so, at the end of Case 2 (§5).

Example 2. Let X be a CW-complex and the X_i subcomplexes with union X. Then X is the homotopy direct limit of $\{X_i\}$.

Milnor's proof: p induces isomorphisms of homotopy groups in all dimensions, so Whitehead's theorem applies. Both halves rest on facts from outside the book.

Facts (CW-complexes). (a) A compact subset of a CW-complex lies in a finite subcomplex (Hatcher, *Algebraic Topology*, Proposition A.1). (b) If X and Y are CW-complexes and Y is locally compact, or both have countably many cells, the product topology on $X \times Y$ is the CW topology of the product cell structure (Hatcher, Theorem A.6); a subcomplex is closed and is a CW-complex in the subspace topology. (c) *Whitehead's theorem*: a map between connected CW-complexes inducing isomorphisms on all homotopy groups π_k, $k \geq 1$, is a homotopy equivalence (J. H. C. Whitehead, *Combinatorial homotopy I*, Bull. Amer. Math. Soc. 55 (1949), 213–245; Hatcher, Theorem 4.5). Applied to each path component, it says that a *weak homotopy equivalence* between CW-complexes (a bijection on path components and isomorphisms on every π_k at every base point) is a homotopy equivalence. Milnor (p. 24) quotes the version for spaces dominated by CW-complexes. (d) *Cellular approximation*: every map between CW-complexes is homotopic to a cellular map (Hatcher, Theorem 4.8). (e) *CW approximation*: for every space Z there are a CW-complex Z' and a weak homotopy equivalence $Z' \to Z$ (Hatcher, Proposition 4.13).

Lemma 2.3. If every compact subset of X lies in some X_n, then $p : X_\Sigma \to X$ is a weak homotopy equivalence.

Proof. A compact subset of X_Σ has bounded second coordinate. *Base points.* For $(x_0, \tau_0) \in X_\Sigma$ and an integer $m \geq \tau_0$, the path $\tau \mapsto (x_0, \tau)$, $\tau_0 \leq \tau \leq m$, lies in X_Σ (T1) and projects to a constant path, so changing base point along it commutes with p_*; we may take base points (x_0, m) with $x_0 \in X_m$. *Surjectivity on π_k.* A map $\alpha : (S^k, *) \to (X, x_0)$ has compact image, inside some X_N with $N \geq m$. Then (α, N) maps into $X_N \times \{N\} \subseteq X_\Sigma$, is based at (x_0, N) and lies over α; move its base point back to (x_0, m) along $\tau \mapsto (x_0, \tau)$. *Injectivity.* Let $\beta = (\beta_1, \beta_2) : (S^k, *) \to (X_\Sigma, (x_0, m))$ and let $G : S^k \times [0,1] \to X$ be a null-homotopy of $p\beta = \beta_1$ relative to the base point. Choose $N \geq m$ with $G(S^k \times [0,1]) \subseteq X_N$ and $N \geq \max \beta_2$. The homotopy $(\beta_1, \beta_2 + s(N - \beta_2))$ stays in X_Σ (T1), moves the base point along $\tau \mapsto (x_0, \tau)$, and ends at (β_1, N), which is null-homotopic relative to the base point through (G, N). So $[\beta]$ corresponds to the trivial class under a change of base point, and is trivial. Path components ($k = 0$) are handled in the same way. □

Proof of Example 2. Every cell of X lies in some X_i, and the X_i increase, so by Fact (a) every compact subset of X lies in some X_n, and p is a weak homotopy equivalence by Lemma 2.3. Give $\mathbb{R}$ the cell structure with vertices at the integers. By Fact (b), $X \times \mathbb{R}$ is a CW-complex in the product topology, and X_Σ, the union of the subcomplexes $X_n \times [n, n+1]$, is a subcomplex, hence a CW-complex in the subspace topology. Whitehead's theorem (Fact (c)) finishes the proof. □

Example 3. The unit interval $[0,1]$ is not the homotopy direct limit of the closed subsets $X_i = \{0\} \cup [\frac{1}{i+1}, 1]$. (Milnor writes $[0] \cup [1/i, 1]$, indexing from $i = 1$.)

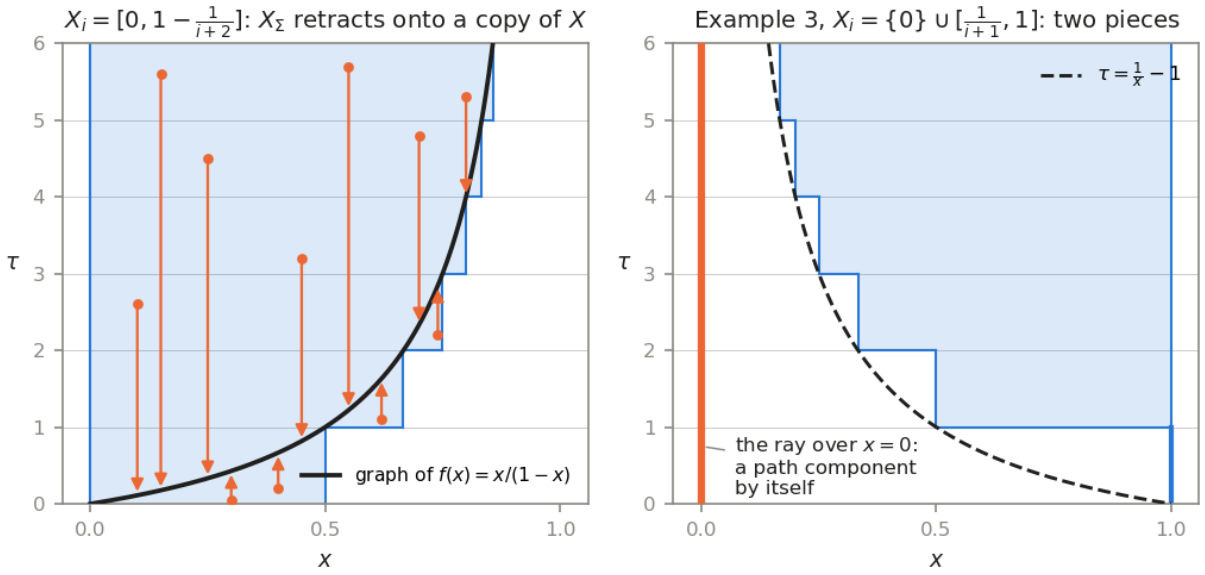
Proof. A homotopy equivalence induces a bijection on path components, and $[0,1]$ is path connected; we show that X_Σ is not. Let $c(s) = (x(s), \tau(s))$, $0 \leq s \leq 1$, be a path in X_Σ with $x(0) = 0$, and suppose $x(s_1) > 0$ for some s_1. Let s^* be the last $s \leq s_1$ with $x(s) = 0$, and $T = \max \tau$. For $s^* < s \leq s_1$, $x(s) > 0$, so $x(s) \in [\frac{1}{\lfloor \tau(s) \rfloor + 1}, 1]$ and $x(s) \geq \frac{1}{T+1}$; letting $s \downarrow s^*$ gives $x(s^*) \geq \frac{1}{T+1} > 0$, a contradiction. So the ray $\{0\} \times [0, \infty)$ is a path component of X_Σ, and X_Σ has at least two. □

Here 0 lies in every X_i but in the interior of none: the hypothesis of Example 1 fails at a single point, and the telescope remembers it.

An interactive look at the telescope

The figure draws telescopes of subsets of an interval in the plane, with x horizontal and τ vertical. In the first example $X = [0, 1)$ and $X_i = [0, 1 - \frac{1}{i+2}]$: every point is interior to some X_i, and X_Σ is the shaded staircase. The curve is the graph of $f(x) = x/(1-x)$, which satisfies $f \geq i + 1$ off X_i, so Lemma 2.1 applies. Drag the slider s to run the deformation H_s: the sample points of X_Σ slide vertically onto the graph, a copy of X. In the second example (Example 3), $X = [0, 1]$ and $X_i = \{0\} \cup [\frac{1}{i+1}, 1]$. The ray over $x = 0$ never touches the staircase. A function whose graph lay in X_Σ would need $f(x) \geq \frac{1}{x} - 1$ for $x > 0$ (dashed curve), so it cannot be continuous at 0. Here the slider runs the push-up deformation of (T2) instead, and the gap never closes. The buttons switch examples; the second slider sets the number of levels drawn.

(In the printed version this figure is a still image; the interactive version is in the web edition.)



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3. Theorem A and the alternative proof of Theorem 17.3 (Milnor p. 150)

Theorem A. *(checked by machine: VA.12, in a narrower form; VA.15, as stated)*[1] Suppose that X is the homotopy direct limit of $\{X_i\}$ and Y is the homotopy direct limit of $\{Y_i\}$. Let $f : X \to Y$ be a

[1]Checked by machine: the result has been verified in the Lean proof assistant. The numbers are those of the verified edition and of the comparison, which give the checked statement in full. *As stated*: the same statement. *Restated*: an equivalent statement in other terms. *With its hypotheses made explicit*: a hypothesis the book uses silently is stated. *In a stronger form*: the checked statement says more than the book's. *In a narrower form*: it covers less, a special setting or a weaker conclusion. *With a sign corrected*: the book's statement has a slip. *Not checked*: no part of the result has been verified. The section "Checked by machine" at the end of the chapter says which forms were checked, and covers the results that carry no such note.

map which carries each X_i into Y_i by a homotopy equivalence $f_i = f|X_i : X_i \to Y_i$. Then f itself is a homotopy equivalence.

No closedness is assumed. Milnor's proof (§§4–5) uses closed Y_i, and the general case is reduced to it at the end of §5.

The alternative final step of Theorem 17.3. In the proof of Theorem 17.3 (Chapter 10) one has non-critical values $a_0 < a_1 < \cdots$ of E, CW-complexes $K_0 \subseteq K_1 \subseteq \cdots$, each a subcomplex of the next, and homotopy equivalences $f_i : \Omega^{a_i} \to K_i$, each extending the previous one (Milnor p. 23). Let $K = \bigcup_i K_i$, a CW-complex with the direct limit topology. To apply Theorem A one needs the following.

- $\Omega = \bigcup_i \Omega^{a_i}$, that is, $a_i \to \infty$. This is automatic if E has infinitely many critical values, since each Ω^a contains only finitely many geodesics (Milnor §16); otherwise the sequence must be continued beyond the last critical value.
- Each $\omega \in \Omega$ lies in the interior of some Ω^{a_i}. The energy is continuous for Milnor's metric on Ω (§16: he adds the term $[\int_0^1 (\frac{ds}{dt} - \frac{ds'}{dt})^2 dt]^{1/2}$ to the metric for exactly this purpose), so $E^{-1}[0, a_i)$ is an open neighbourhood of ω in Ω^{a_i} once $a_i > E(\omega)$. The same continuity makes each Ω^{a_i} closed.
- Ω is paracompact, being a metric space (A. H. Stone's theorem; Munkres, Theorem 41.4).
- The union map $f : \Omega \to K$, $f|\Omega^{a_i} = f_i$, is well defined because each f_{i+1} extends f_i, and continuous because it agrees with the continuous f_i on the open sets $E^{-1}[0, a_i)$, which cover Ω.

Then Ω is the homotopy direct limit of $\{\Omega^{a_i}\}$ (Example 1), K is that of $\{K_i\}$ (Example 2), and Theorem A says that $f : \Omega \to K$ is a homotopy equivalence. Milnor's first proof (p. 95) instead uses that Ω has the homotopy type of a CW-complex (his Corollary 17.2, resting on his paper *On spaces having the homotopy type of a CW-complex*, Trans. Amer. Math. Soc. 90 (1959), 272–280), because Whitehead's theorem upgrades the weak equivalence $\Omega \to K$ to a homotopy equivalence only for such spaces. Being weakly equivalent to a CW-complex is no help by itself: by CW approximation (Fact (e)) every space is. Theorem A uses Whitehead's theorem only for CW-complexes, inside Example 2.

Reduction to the telescopes. Define $f_\Sigma : X_\Sigma \to Y_\Sigma$ by $f_\Sigma(x, \tau) = (f(x), \tau)$; it lands in Y_Σ because $f(X_{\lfloor\tau\rfloor}) \subseteq Y_{\lfloor\tau\rfloor}$. Milnor calls it "clearly sufficient" to prove that f_Σ is a homotopy equivalence. Indeed $p_Y f_\Sigma = f p_X$; if s_X is a homotopy inverse of p_X, then $f \simeq f p_X s_X = p_Y f_\Sigma s_X$, a composite of homotopy equivalences, and a map homotopic to a homotopy equivalence is one.

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4. Case 1: maps homotopic to the identity on every level (Milnor pp. 150–152)

Case 1. Suppose $X_i = Y_i$ for all i and each $f_i : X_i \to X_i$ is homotopic to the identity of X_i. We prove that f_Σ is a homotopy equivalence. (This part does not use that X is a homotopy direct limit.) Throughout this section the X_i are closed: every map below is defined strip by strip, and (T3) makes it continuous. The blockquote in §1 shows what goes wrong otherwise; §5 needs Case 1 only for closed sets.

Remark (Milnor). Under these conditions one might conjecture that f_Σ is actually homotopic to the identity; Milnor says that "counterexamples can be given".

Milnor gives none. Here is one; in later language it is built from a *phantom map*, one that is null-homotopic on every finite subcomplex but not null-homotopic.

A counterexample. Let T be the telescope of $S^1 \xrightarrow{z^2} S^1 \xrightarrow{z^2} \cdots$: cylinders $S^1 \times [n, n+1]$, $n \geq 0$, with the top circle of the n-th glued to the bottom circle of the $(n+1)$-st by $z \mapsto z^2$. Its cells are the vertices $v_n = (1, n)$, the edges a_n (the circle at level n) and $c_n = \{1\} \times (n, n+1)$, and the 2-cells $D_n = (S^1 \setminus 1) \times (n, n+1)$, with $\partial D_n = a_n - 2a_{n+1}$: the bottom of the square runs once around a_n, the top twice around a_{n+1}, and the two sides cancel. Let T_n be the subcomplex made of the first n cylinders; it deformation retracts onto a circle.

(i) $H^2(T; \mathbb{Z}) \neq 0$. There are no 3-cells, so $H^2(T; \mathbb{Z})$ is the group of integer sequences (b_n) (one value per D_n) modulo those of the form $b_n = \varphi_n - 2\varphi_{n+1}$. For $b = (1, 0, 1, 0, \dots)$ such integers would satisfy $\varphi_0 = \sum_{k<N} b_k 2^k + 2^N \varphi_N$ for every N; for even N, $3\sum_{k<N} b_k 2^k = 4^{N/2} - 1$, so $3\varphi_0 + 1$ would be divisible by every power of 2, impossible as $3\varphi_0 + 1 \neq 0$.

(ii) Let $\mathbb{CP}^n$ be the nonzero complex polynomials of degree $\leq n$ in one variable, up to scalars, and $\mathbb{CP}^\infty = \bigcup_n \mathbb{CP}^n$ with its usual cells. Multiplication of polynomials makes $\mathbb{CP}^\infty$ a commutative monoid with unit $e = [1]$ (a 0-cell), continuous because the product topology on $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ is the CW topology (Fact (b)). It is simply connected, and its 2-skeleton $\mathbb{CP}^1$ is a 2-sphere. Collapse the 1-skeleton of T to get $T \to \bigvee_n S^2_n$, and follow it by the map that is a homeomorphism $S^2_n \to \mathbb{CP}^1$ (wedge point to e) when $b_n = 1$ and constant when $b_n = 0$. The resulting $g : T \to \mathbb{CP}^\infty$ pulls a generator of $H^2(\mathbb{CP}^\infty; \mathbb{Z})$ back to the class of the cellular cocycle b (cellular cohomology and its naturality: Hatcher, §3.1), which is nonzero, so g is not null-homotopic. But each $g|T_n$ is, because T_n retracts by deformation onto a circle and $\pi_1(\mathbb{CP}^\infty) = 0$.

(iii) Let $W = T \times \mathbb{CP}^\infty$ (a CW-complex by Fact (b)), $W_n = T_n \times \mathbb{CP}^\infty$, and $f(x, z) = (x,\ g(x)z)$. Then $f(W_n) \subseteq W_n$, and if g_u is a null-homotopy of $g|T_n$ ending at e, $(x, z) \mapsto (x, g_u(x)z)$ is a homotopy in W_n from $f|W_n$ to the identity: Case 1 applies. But $f \not\simeq \mathrm{id}_W$: restricting a homotopy from f to id to $T \times \{e\}$ and projecting to $\mathbb{CP}^\infty$ would give a homotopy from g to the constant map e. Hence $f_\Sigma \not\simeq$ id: W is a homotopy direct limit (Example 2), and if $f_\Sigma \simeq$ id, then $f \simeq fps = pf_\Sigma s \simeq ps \simeq$ id for a homotopy inverse s of p.

Milnor's proof of Case 1. For each n let $h^n_u : X_n \to X_n$, $0 \leq u \leq 1$, be a homotopy with $h^n_0 = f_n$ and $h^n_1 = \mathrm{id}$. On the strip $X_n \times [n, n+1]$ define

$$h_u(x, n+t) = \begin{cases} (h^n_u(x),\ n + 2t) & 0 \leq t \leq \frac{1}{2}, \\ (h^n_{(3-4t)u}(x),\ n+1) & \frac{1}{2} \leq t \leq \frac{3}{4}, \\ (h^{n+1}_{(4t-3)u}(x),\ n+1) & \frac{3}{4} \leq t \leq 1. \end{cases}$$

Milnor's text skips… the correct index in the third line: it prints $h^{n+1}_{(4-3t)u}$, which at $t = \frac{3}{4}$ would be $h^{n+1}_{7u/4}$. It must be $h^{n+1}_{(4t-3)u}$, running from $h^{n+1}_0 = f_{n+1}$ to h^{n+1}_u.

Well defined and continuous. At $t = \frac{1}{2}$ both lines give $(h^n_u(x), n+1)$. At $t = \frac{3}{4}$ both give $(f(x), n+1)$, since $h^n_0 = f_n$, $h^{n+1}_0 = f_{n+1}$ and $f_{n+1}|X_n = f_n$. At $t = 1$ the third line gives $(h^{n+1}_u(x), n+1)$, the value of the first line on the next strip at $t = 0$. The values lie in X_Σ, each formula is continuous in (x, t, u), and (T3) gives continuity of $(z, u) \mapsto h_u(z)$.

$h_0 \simeq f_\Sigma$ (Milnor: "clearly"). $h_0(x, n+t) = (f(x), n + \min(2t, 1)) = f_\Sigma(r(x, n+t))$ with

$r(x, n+t) = (x, n + \min(2t, 1))$, and $r \simeq \mathrm{id}$ through $(x, n + (1-s)t + s\min(2t, 1))$, which only raises τ inside the strip.

The map h_1. It satisfies

$$h_1(x, n+t) = (x, n+2t) \ \ (0 \le t \le \tfrac{1}{2}), \qquad h_1(x, n+t) \in X_{n+1} \times \{n+1\} \ \ (\tfrac{1}{2} \le t \le 1),$$

and $h_1(y, m) = (h_1^m(y), m) = (y, m)$ at every integer level m. Milnor shows that any such map is a homotopy equivalence, with homotopy inverse

$$g(x, n+t) = \begin{cases} (x, n+2t) & 0 \le t \le \frac{1}{2}, \\ h_1(x, n + \frac{3}{2} - t) & \frac{1}{2} \le t \le 1, \end{cases}$$

well defined since $h_1(x, n+\frac{1}{2}) = h_1(x, n+1) = (x, n+1)$, continuous by (T3). Like h_1, g fixes the points at integer levels.

$h_1 g \simeq \mathrm{id}$. Since h_1 fixes the integer levels and maps $(x, n+s)$, $s \ge \frac{1}{2}$, to level $n+1$,

$$h_1 g(x, n+t) = \begin{cases} (x, n+4t) & 0 \le t \le \frac{1}{4}, \\ h_1(x, n+2t) & \frac{1}{4} \le t \le \frac{1}{2}, \\ h_1(x, n + \frac{3}{2} - t) & \frac{1}{2} \le t \le 1. \end{cases}$$

On $[\frac{1}{4}, 1]$ this runs forwards and then backwards along the loop $s \mapsto h_1(x, n+s)$, $\frac{1}{2} \le s \le 1$, which starts and ends at $(x, n+1)$. Milnor's homotopy cuts the excursion short: for $0 \le u \le \frac{1}{2}$,

$$H_u(x, n+t) = \begin{cases} h_1 g(x, n+t) & 0 \le t \le \frac{1-u}{2} \ \text{ or } \ \frac{1}{2} + u \le t \le 1, \\ h_1(x, n+1-u) & \frac{1-u}{2} \le t \le \frac{1}{2} + u, \end{cases}$$

well defined because $h_1 g(x, n + \frac{1-u}{2}) = h_1 g(x, n + \frac{1}{2} + u) = h_1(x, n+1-u)$. Then $H_0 = h_1 g$, and $H_{1/2}(x, n+t) = (x, n + \min(4t, 1))$ because $h_1(x, n+\frac{1}{2}) = (x, n+1)$; this is homotopic to the identity by raising τ as before.

$g h_1 \simeq \mathrm{id}$ (Milnor: "a completely analogous argument"). Since g fixes the integer levels,

$$g h_1(x, n+t) = \begin{cases} (x, n+4t) & 0 \le t \le \frac{1}{4}, \\ h_1(x, n + \frac{3}{2} - 2t) & \frac{1}{4} \le t \le \frac{1}{2}, \\ h_1(x, n+t) & \frac{1}{2} \le t \le 1, \end{cases}$$

which runs backwards along the same loop and then forwards. With $\psi(t) = \frac{3}{2} - 2t$ on $[\frac{1}{4}, \frac{1}{2}]$ and $\psi(t) = t$ on $[\frac{1}{2}, 1]$, the homotopy

$$K_u(x, n+t) = \begin{cases} (x, n+4t) & 0 \le t \le \frac{1}{4}, \\ h_1(x, n + \max(\psi(t), \frac{1}{2} + u)) & \frac{1}{4} \le t \le 1, \end{cases} \qquad 0 \le u \le \tfrac{1}{2},$$

is well defined ($\psi = 1$ at $t = \frac{1}{4}$ and at $t = 1$) and joins $K_0 = g h_1$ to $K_{1/2}(x, n+t) = (x, n + \min(4t, 1)) \simeq \mathrm{id}$.

So h_1 is a homotopy equivalence, $f_\Sigma \simeq h_0 \simeq h_1$ through the h_u, and f_Σ is a homotopy equivalence. □

5. Case 2: the general case (Milnor pp. 152–153)

Let X, Y be as in Theorem A, and suppose first that the Y_i are closed; this assumption is removed at the end of the section. For each n let $g_n : Y_n \to X_n$ be a homotopy inverse of f_n, and $i_n : X_n \subseteq X_{n+1}$, $j_n : Y_n \subseteq Y_{n+1}$ the inclusions. The square formed by g_n, g_{n+1}, i_n, j_n commutes up to homotopy:

$$i_n g_n \simeq g_{n+1} f_{n+1} i_n g_n = g_{n+1} j_n f_n g_n \simeq g_{n+1} j_n,$$

using $g_{n+1} f_{n+1} \simeq \mathrm{id}$, $f_{n+1} i_n = j_n f_n$ and $f_n g_n \simeq \mathrm{id}$. Choose homotopies $h_u^n : Y_n \to X_{n+1}$ with $h_0^n = i_n g_n$ and $h_1^n = g_{n+1} j_n$, and define $G : Y_\Sigma \to X_\Sigma$ by

$$G(y, n+t) = \begin{cases} (g_n(y),\ n+2t) & 0 \le t \le \frac{1}{2}, \\ (h_{2t-1}^n(y),\ n+1) & \frac{1}{2} \le t \le 1. \end{cases}$$

It is well defined ($h_0^n = g_n$ at $t = \frac{1}{2}$; at $t = 1$, $h_1^n = g_{n+1}$ matches the next strip), takes values in X_Σ, and is continuous by (T3).

Gf_Σ is a homotopy equivalence. $Gf_\Sigma(x, n+t)$ is $(g_n f_n(x), n+2t)$ or $(h_{2t-1}^n(f(x)), n+1)$, so Gf_Σ maps each truncation X_Σ^m into itself, and on $X_m \times \{m\}$ it is $(x, m) \mapsto (g_m f_m(x), m)$, homotopic in $X_m \times \{m\}$ to the identity. Let φ be the restriction of Gf_Σ to X_Σ^m, $r : X_\Sigma^m \to X_m \times \{m\}$ the retraction of (T2) and ι the inclusion. With all homotopies inside X_Σ^m,

$$\varphi \simeq \varphi\, \iota r = (\varphi|_{X_m \times \{m\}})\, r \simeq \iota r \simeq \mathrm{id}.$$

So Gf_Σ satisfies the hypotheses of Case 1 for the space X_Σ and its subsets X_Σ^m, which are closed whether or not the X_i are (they are cut out by $\tau \le m$), and Case 1 says that the induced map $(Gf_\Sigma)_\Sigma$ of the telescope of $\{X_\Sigma^m\}$ is a homotopy equivalence. As X_Σ is the homotopy direct limit of $\{X_\Sigma^m\}$ (Lemma 2.2), the reduction in §3 shows that Gf_Σ itself is a homotopy equivalence.

> **Milnor's text skips…** this last step. Case 1 is about the induced map of telescopes; to come back to Gf_Σ one needs X_Σ to be the homotopy direct limit of its truncations. Example 1 does not give this (X_Σ need not be paracompact), but Lemma 2.2 does.

Conclusion. If k is a homotopy inverse of Gf_Σ, then $(kG)f_\Sigma \simeq \mathrm{id}$: f_Σ has a left homotopy inverse. Symmetrically, $f_\Sigma G : Y_\Sigma \to Y_\Sigma$, given by $(f_n g_n(y), n+2t)$ and $(f(h_{2t-1}^n(y)), n+1)$, preserves the truncations Y_Σ^m and is $f_m g_m \simeq \mathrm{id}$ on $Y_m \times \{m\}$; so it is a homotopy equivalence, with inverse k' say, and $f_\Sigma(Gk') \simeq \mathrm{id}$: f_Σ has a right homotopy inverse. By the Assertion on Milnor's p. 22 (Chapter 2: a map with a left and a right homotopy inverse is a homotopy equivalence), f_Σ is a homotopy equivalence, and by §3 so is f. This proves Theorem A when the Y_i are closed. □

Theorem A in general. Now let the X_i and Y_i be arbitrary. Give the set X_Σ the finest topology in which every strip $X_n \times [n, n+1] \to X_\Sigma$ is continuous, and call the result T_X: it is the *mapping telescope* of the inclusions $X_n \subseteq X_{n+1}$, the strips glued along their common levels. A map out of T_X is continuous as soon as it is continuous on every strip, and so is a map out of $[0,1] \times T_X$, since the product of a quotient map with the compact space $[0,1]$ is a quotient map: this is (T3) without its hypothesis. The identity $T_X \to X_\Sigma$ is continuous, and a homeomorphism when the X_i are closed, by (T3). The truncations $T_X^n = \{\tau \le n\}$ are closed in T_X, and T_X is their homotopy direct limit, by Lemma 2.1 with $f(x, \tau) = \tau + 1$. (T2) holds in T_X with the same deformation $(x, \tau) \mapsto (x, \tau + s(n - \tau))$: on $[0,1]$ times a strip, the moving point crosses finitely many strips, on closed pieces, so the deformation is continuous; and $(x, \tau) \mapsto x$ is continuous from T_X^n to X_n,

which carries the topology of X. So $\kappa_n^X : x \mapsto (x, n)$ is a homotopy equivalence $X_n \to T_X^n$, with homotopy inverse $q_n^X(x, \tau) = x$.

The identity $c_X : T_X \to X_\Sigma$ carries T_X^n into the closed set X_Σ^n, and there it is homotopic to $c_X \kappa_n^X q_n^X$, a composite of homotopy equivalences by (T2). By Theorem A for closed Y_i, with Lemmas 2.1 and 2.2, c_X is a homotopy equivalence, and so is $p_X c_X : T_X \to X$, $(x, \tau) \mapsto x$, since X is the homotopy direct limit of $\{X_i\}$; likewise for Y. The map $f_T : T_X \to T_Y$, $f_T(x, \tau) = (f(x), \tau)$, is continuous on every strip, carries T_X^n into the closed set T_Y^n, and there it is homotopic to $f_T \kappa_n^X q_n^X = \kappa_n^Y f_n q_n^X$, a homotopy equivalence. By Theorem A for closed Y_i again, f_T is a homotopy equivalence. Since $f(p_X c_X) = (p_Y c_Y) f_T$, the argument of §3 shows that f is a homotopy equivalence. This proves Theorem A as Milnor states it. □

So the closedness that Milnor's proof needs is a feature of the telescope X_Σ, not of the theorem: the proof is the same, run on the telescopes T_X and T_Y, where pasting is automatic, and on their truncations, which are closed.

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6. The Corollary (Milnor p. 153)

Corollary. Suppose that X is the homotopy direct limit of $\{X_i\}$. If each X_i has the homotopy type of a CW-complex, then X itself has the homotopy type of a CW-complex.

"The proof is not difficult," says Milnor, and gives none. One cannot just take a union of CW-complexes $K_i \simeq X_i$, since nothing makes K_i a subcomplex of K_{i+1}; a telescope of the K_i does the job.

Proof. Choose CW-complexes K_i and homotopy equivalences $e_i : X_i \to K_i$ with homotopy inverses e_i'. By cellular approximation (Fact (d)) the maps $e_{i+1} \iota_i e_i' : K_i \to K_{i+1}$, with $\iota_i : X_i \subseteq X_{i+1}$ the inclusion, are homotopic to cellular maps c_i. Let T be the mapping telescope

$$T = \Big(\bigsqcup_i K_i \times [i, i+1] \Big) \Big/ \big((k, i+1) \sim (c_i(k), i+1) \big),$$

a CW-complex because the c_i are cellular (Hatcher, §3.F), with subcomplexes T_n (the first n blocks together with $K_n \times \{n\}$) whose union is T. Collapsing the blocks one at a time, $(k, s) \mapsto (k, s + u(i + 1 - s))$ on the i-th, deforms T_n onto $K_n \times \{n\}$, so the inclusion $\kappa_n : K_n \to T_n$ at level n is a homotopy equivalence; and T is the homotopy direct limit of $\{T_n\}$ by Example 2. Since $c_i e_i \simeq e_{i+1} \iota_i e_i' e_i \simeq e_{i+1} \iota_i$, there are homotopies $H_u^i : X_i \to K_{i+1}$ from $c_i e_i$ to $e_{i+1} \iota_i$. On the mapping telescope T_X of the inclusions (§5: the set X_Σ, with the topology in which a map is continuous as soon as it is continuous on every strip) define $F : T_X \to T$ as G was defined in Case 2, writing $[k, \tau]$ for the class of (k, τ) in T:

$$F(x, n+t) = \begin{cases} [e_n(x),\ n + 2t] & 0 \le t \le \tfrac{1}{2}, \\ [H_{2t-1}^n(x),\ n+1] & \tfrac{1}{2} \le t \le 1. \end{cases}$$

At $t = \frac{1}{2}$, $[e_n(x), n+1] = [c_n e_n(x), n+1] = [H_0^n(x), n+1]$; at $t = 1$ the value $[e_{n+1}(x), n+1]$ matches the next strip. So F is continuous, with no closedness needed; it maps T_X^n into T_n, and $F(x, n) = \kappa_n e_n(x)$. With κ_n^X and q_n^X as in §5, $F | T_X^n \simeq F \kappa_n^X q_n^X = \kappa_n e_n q_n^X$, a composite of homotopy equivalences. So F carries each T_X^n into T_n by a homotopy equivalence. As T_X is the homotopy direct limit of $\{T_X^n\}$ (§5) and T that of its subcomplexes T_n, which are closed, Theorem A shows that F is a homotopy equivalence. Hence $X \simeq T_X \simeq T$ (§5). □

7. Application: the path space of a sphere, level by level

Let $p, q \in S^n$, $n \geq 3$, be at angle $\theta = \frac{\pi}{2}$; they are not conjugate (the conjugate points of p along geodesics of the unit sphere are $\pm p$, Chapter 9). By Chapter 7, §6, the geodesics from p to q are the arcs $\gamma_0, \gamma_1, \gamma_2, \ldots$ of the great circle through them, of lengths $(2k+1)\frac{\pi}{2}$, where γ_k passes k times through p or its antipode. Milnor's energy $\int_0^1 \|\omega'\|^2 dt$ of a constant-speed path is the square of its length, so

$$E(\gamma_k) = (2k+1)^2 \frac{\pi^2}{4} : \qquad 2.47,\ \ 22.21,\ \ 61.69,\ \ 120.90,\ \ 199.86,\ \ldots$$

By Milnor §17 (Chapter 10), γ_k has index $k(n-1)$. Take $a_i = (i+1)^2\pi^2$, that is $9.87,\ 39.48,\ 88.83,\ 157.91, \ldots$, which lies strictly between $E(\gamma_i)$ and $E(\gamma_{i+1})$ because $(2i+1)^2 < (2i+2)^2 < (2i+3)^2$. Then Ω^{a_i} contains exactly the geodesics $\gamma_0, \ldots, \gamma_i$, and the proof of Theorem 17.3 gives $\Omega^{a_i} \simeq K_i$, a CW-complex with one cell in each dimension $0, n-1, 2(n-1), \ldots, i(n-1)$. For $n \geq 3$, consecutive cells differ in dimension by $n-1 \geq 2$, so all cellular boundary maps vanish and $H_k(\Omega^{a_i}; \mathbb{Z}) \cong \mathbb{Z}$ for $k = 0, n-1, \ldots, i(n-1)$, and 0 otherwise.

This is where Theorem A enters: with Examples 1 and 2, it gives $\Omega(S^n; p, q) \simeq K = \bigcup_i K_i$, a CW-complex with one cell in each dimension divisible by $n-1$, hence

$$H_k(\Omega(S^n; p, q); \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & (n-1) \mid k, \\ 0 & \text{otherwise,} \end{cases} \qquad \sum_k \operatorname{rank} H_k \, t^k = \frac{1}{1 - t^{n-1}}.$$

For S^3: one generator in each even degree.

The finite levels also say something quantitative. A singular cycle representing a generator of $H_{k(n-1)}(\Omega)$ must contain a path of energy at least $E(\gamma_k)$: otherwise, its image being compact, it would lie in Ω^a for some $a \in (E(\gamma_{k-1}), E(\gamma_k))$, which is homotopy equivalent to a complex with no cells above dimension $(k-1)(n-1)$, so the cycle would bound there. On S^3 with $\theta = \frac{\pi}{2}$, any family of paths representing the generator of H_4 contains a path of energy at least $\frac{25\pi^2}{4} \approx 61.7$, the energy of the geodesic γ_2 that wraps once past p and once past its antipode.

8. Summary of what was silently assumed, now made explicit

Where	Milnor says	What is actually needed
p. 149	X_Σ, "topologized as a subset of $X \times \mathbb{R}$"	$(x, \tau) \in X_\Sigma \iff x \in X_{\lfloor \tau \rfloor}$; upward closure (T1); truncations (T2)
p. 149	Example 1: "using a partition of unity"	$f = \sum_i i\,\varphi_i$; paracompact includes Hausdorff
p. 149	"which is clearly a deformation retract"	the straight-line homotopy in τ stays in X_Σ (Lemma 2.1)

Where	Milnor says	What is actually needed
p. 149	Example 2: "p induces isomorphisms of homotopy groups"	compact sets lie in finite subcomplexes; base points (Lemma 2.3)
p. 149	"it follows from Whitehead's theorem"	X_Σ is a CW-complex, as $X \times \mathbb{R}$ has the CW topology
p. 150	Example 3 (no proof)	X_Σ has two path components
p. 150	Theorem A	true as stated; the proof pastes maps strip by strip (T3), so Case 1 needs the X_i closed and the proof the Y_i; mapping telescopes remove both (§5)
p. 150	the alternative proof of 17.3	$a_i \to \infty$; E continuous; Ω metric, so paracompact; f_{i+1} extends f_i
p. 150	reducing to f_Σ is "clearly sufficient"	$p_Y f_\Sigma = f p_X$
p. 150	"counterexamples can be given"	none given; a phantom-map example in §4
p. 151	third line of h_u: $h^{n+1}_{(4-3t)u}$	possible misprint for $h^{n+1}_{(4t-3)u}$
p. 151	h_0 "clearly homotopic to f_Σ"	$h_0 = f_\Sigma \circ r$, r a reparametrization of τ
p. 151	$h_1 g(x, n+t) = h_1(x, n+\frac{3}{2}-t)$ for $t \geq \frac{1}{2}$	h_1 is the identity at integer levels
p. 152	"gh_1 is homotopic to the identity" by "a completely analogous argument"	the homotopy K_u
p. 152	"apply Case 1 to the sequence $\{X^n_\Sigma\}$"	the truncations are closed, and X_Σ is their homotopy direct limit (Lemma 2.2)
p. 153	"compare page 22"	left and right homotopy inverses give a homotopy equivalence
p. 153	Corollary: "The proof is not difficult"	mapping telescope of cellular maps, and Theorem A (§6)

Checked by machine. The Appendix has been verified in the Lean proof assistant, for arbitrary topological spaces: telescopes, homotopy direct limits, (T1)–(T3) and Lemmas 2.1–2.2; Example 1, and its form for the sublevel sets of a continuous function, which needs no paracompactness; Example 3; the reduction to f_Σ; and both cases of Milnor's proof of Theorem A, with the corrected $(4t-3)u$. The machine also settled the question of closedness: Case 1 as written needs the X_i closed, Milnor's proof of Theorem A needs only the Y_i closed, and Theorem A holds as Milnor states it, by his proof run on mapping telescopes. The Corollary is checked with a mapping telescope of the K_i in place of a CW-complex, and so is the topological part of Theorem 17.3, for a space exhausted by sublevel sets with compatible towers of cells. Not checked: Example 2, which needs Whitehead's theorem, and Lemma 2.3; a counterexample for the Remark; the CW structure

of a mapping telescope, which needs cellular approximation; and the application. See the verified edition.